Analytical and Physico-Mathematical Refutation of Thermodynamic Constraints on the Simulation Hypothesis

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Аннотация

This work presents a rigorous mathematical dismantling of the thermodynamic arguments of Franco Vazza against the simulation hypothesis. From the standpoint of quantum information theory, noncommutative geometry, and the AdS/CFT holographic principle, we demonstrate the methodological inconsistency of extrapolating macroscopic thermodynamic laws to the fundamental quantum-informational substrate of reality. Special attention is devoted to formalizing the epistemological barrier of the carbon-based observer (the "Wall of Cognition") and to justifying the transition to photonic computational substrates. The paper synthesizes the AdS/CFT duality apparatus with the theory of ultragraphs, allowing us to describe the assembly of continuous spacetime from discrete informational blocks of the parent matrix. Additionally, we introduce the formalism of von Neumann type III algebras to rigorously prove the infinite heat capacity of the CFT boundary and provide specific experimental proposals for detecting the "observer codec" via quantum optics.

Introduction and Problem Statement

In contemporary theoretical physics, the debate on the legitimacy of the simulation hypothesis (the informational nature of the observed continuum) has shifted from qualitative philosophical discourse to the domain of rigorous quantum infodynamics. The resonant work of the Italian astrophysicist Franco Vazza [1] posits the thermodynamic and gravitational impossibility of modeling our Universe (or its representative macroscopic part, such as planet Earth) within a parent continuum governed by the same physical laws.

Vazza's central postulate: Any computational carrier (the simulator's processor) is inevitably associated with energy dissipation during information rewriting and erasure (Landauer's principle) and is limited by the maximum information packing density (Bekenstein bound). Vazza's calculations show that continuous rendering of even local subquantum processes (e.g., trajectories of high-energy neutrinos detected by the IceCube observatory) requires computational power and energy expenditure that would lead to the immediate gravitational collapse of the Creator's processor into a black hole due to critical energy density in a finite volume.

The aim of this analysis is to prove the methodological unsoundness of F. Vazza's calculations, relying on the formalism of noncommutative geometry and the theory of ultragraphs developed in the works of D. N. Boyko [2, 3, 4, 5, 6], as well as the formal model of the "biochemical censorship of the observer" [2].

1 Refutation of the Landauer Dissipative Mechanism in the Microworld

1.1 The Mathematical Dead End of the Classical Bit

Vazza bases his calculations on the classical representation of information as a sequence of discrete bit erasure/write operations, where the minimum heat dissipation per bit is given by:

$$E_{\text{minimal}} = k_B T \ln 2, \tag{1}$$

where k_B is Boltzmann's constant, T is the local system temperature.

Counterargument: At the fundamental sub-Planckian level, the notion of a classical dissipative bit is an illegitimate idealization. True quantum reality is described by the evolution of pure states in Hilbert space governed by the Schrödinger equation:

$$i\hbar \frac{\partial \rho_{\text{total}}}{\partial t} = [H, \rho_{\text{total}}]. \tag{2}$$

The dynamics of a closed quantum system is strictly unitary and reversible. According to the theorems of quantum information theory (including the no-cloning theorem), information in such a system is never lost, duplicated, or erased.

As shown in [5], the noncommutative geometry of distributed computing systems demonstrates that fundamental phase-state changes do not require thermodynamic data erasure. Consequently:

$$\Delta S_{\text{total}} = 0 \quad \Longrightarrow \quad Q_{\text{dissipation}} \equiv 0. \tag{3}$$

1.2 Intermediate Conclusion

Landauer's principle is inapplicable to the fundamental quantum matrix. The heat dissipation calculated by Vazza is identically zero at the base computing level.

2 Deployment of the AdS/CFT Duality Apparatus: Holographic Entropy Sink

To explain why macroscopic processes (including the "energy-intensive" neutrino flux) do not overheat the processor, we invoke the Maldacena duality ($\text{AdS}_{d+1}/\text{CFT}_d$).

2.1 Topology of the Bulk and Boundary

We model our spacetime continuum as a $(d+1)$-dimensional anti-de Sitter space (AdS_{d+1}), whose metric in Poincaré coordinates is:

$$ds^2 = \frac{R^2}{z^2}\left(-dt^2 + d\vec{x}^2 + dz^2\right), \tag{4}$$

where R is the curvature radius, z is the holographic coordinate ($z \in (0, \infty)$). Our macroworld resides in the bulk. The computational complex (the Simulator's processor) is physically localized on the conformal boundary of the continuum (∂AdS as $z \to 0$), where a quantum conformal field theory (CFT_d) devoid of gravity is defined.

The Gubser–Klebanov–Polyakov–Witten correspondence relates the generating functional of bulk fields to the partition function of the quantum code on the boundary:

$$\mathcal{Z}_{\text{Bulk}}[\phi_0(\vec{x},t)] = \left\langle \exp\left(\int d^d x\, \phi_0(\vec{x},t)\mathcal{O}(\vec{x},t)\right)\right\rangle_{\text{CFT}}. \tag{5}$$

In [4], the methodology of reverse reconstruction as a universal ontological basis is formalized, employing quantum nonlocality to construct the informational projection of reality — which organically complements the holographic picture.

2.2 The Ryu–Takayanagi Formula and Scale Dissipation

Local entropy growth measured by macroscopic instruments in the bulk is mapped to boundary geometry via the Ryu–Takayanagi formula:

$$S_A = \frac{\text{Area}(\gamma_A)}{4G_{d+1}}. \tag{6}$$

When projecting bulk energy onto the boundary, the effective Landauer energy density scales as:

$$\lim_{z\to 0} E_{\text{Landauer}} \propto z^{d-1} = 0 \quad (\text{for } d > 1). \tag{7}$$

3 Ultragraph as a Combinatorial Processor: Synthesis of AdS/CFT with Ultragraph Theory

In the classical Maldacena duality, the conformal boundary ($z = 0$) supports a continuous field theory (CFT). However, as shown in [2, 3], from the viewpoint of information theory, continuity on the boundary is an idealization. The fundamental layer of reality is strictly discrete and combinatorial.

Definition 1. *The **Ultragraph of the matrix** is defined as $\mathcal{UG} = (V, E, H)$, where:*

- *V is a set of vertices representing elementary qubits of quantum information (pure phase states);*
- *E are hyperedges connecting arbitrary groups of vertices, reflecting many-particle quantum entanglement;*
- *H are hierarchical operators (ultra-edges) that connect hyperedges themselves, describing meta-entanglement and code restructuring rules.*

Theorem 1. *The conformal field theory (CFT) on the boundary emerges as the low-energy limit (infrared fixed point) of the ultragraph:*

$$\lim_{|V|\to\infty} \mathcal{UG} \simeq CFT_d. \tag{8}$$

3.1 Bulk Assembly via MERA Tensor Networks

To understand how abstract ultragraph connections give rise to physical distances and geometry, we employ the MERA (Multiscale Entanglement Renormalization Ansatz) formalism.

The ultragraph $\mathcal{UG}$ implements a cascade renormalization. The architecture of hierarchical hyperedges H groups lower-level vertices into macronodes. This hierarchical data compression process is mathematically identical to moving inward along the holographic coordinate z:

```
BOUNDARY ULTRAGRAPH (Discrete qubits V, hierarchical links H)
  z = 0 (CFT Boundary)
\ /     \ /     \ /     \ /     \ /     \ /
      z = 1 (Subquantum level)
  \     /         \     /         \     /
              z = 2 (Microparticle level)
    \             /                 /

  [Smooth continuous AdS geometry / Bulk]  z ->  (Macroworld)
```

Each renormalization step of the ultragraph cuts off high-energy (ultraviolet) entanglement. As a result, discrete information blocks, entangled via hyperedges, form spatial quanta. The metric distance between two points in the bulk macroworld is defined as the minimum number of ultragraph hyperedges that must be crossed to connect the information blocks on the boundary.

3.2 Quantum Isomorphism: Ultragraphs and the Amplituhedron

In [3], an isomorphism between ultragraphs and the amplituhedron geometry in the Grassmannian is proven. Let $\mathcal{A}_n$ be the amplituhedron determining the scattering amplitudes of n particles. The differential form of the amplituhedron, which gives the physical probabilities of events, is topologically equivalent to the invariants of the ultragraph:

$$\Omega(\mathcal{A}_n) \equiv \sum_i \text{Index}(H_i \in \mathcal{UG}). \tag{9}$$

Corollary 1. *Since the amplituhedron is a static, rigidly fixed multidimensional geometric object in the positive Grassmannian, the associated ultragraph $\mathcal{UG}$ does not undergo dynamical iterations. The physical "motion" of particles in the AdS bulk is merely a macroscopic sliding along fixed geodesic lines laid out by ultragraph edges. The dynamics of time emerges as an illusion upon decoherence of subsystems within the bulk. Consequently:*

$$\Delta S_{universal} \equiv 0 \quad \Longrightarrow \quad Q_{dissipation} = 0. \tag{10}$$

4 "Biochemical Censorship of the Observer" and Mathematical Formalization of the "Wall of Cognition"

A fundamental question arises: if reality is discrete, dissipativeless, and static at the subquantum level, why does Franco Vazza measure thermodynamic entropy growth, continuous time, and materiality of particles?

The answer lies in the *biochemical censorship of the observer* (the "Wall of Cognition") formulated in [2, 6]. The human brain is a macroscopic carbon-based system with severe bandwidth limitations. In [6], the upper combinatorial limit of abstraction for discrete cognitive systems of human type is formalized.

4.1 Mathematical Reduction of Infinity

The fundamental code on the matrix boundary (CFT_d) possesses UV-completeness. This means that any arbitrarily small region of the boundary contains an infinite number of strongly coupled quantum degrees of freedom (entropy density diverges). The biological brain, constrained by the Bekenstein bound and the speed of electrochemical impulses, cannot physically process pure CFT states.

Mathematically, the perception process is described as a partial trace operation. The brain takes the global density matrix of the boundary ρ_{CFT} and coarse-grains it, discarding phase microstates:

$$\rho_{\mathrm{Observer}} = \mathrm{Tr}_{\text{UV-degrees}}\left(\rho_{\mathrm{CFT}}\right). \tag{11}$$

This process is exactly equivalent to lossy compression. The space we see around us (the AdS bulk) is a compression artifact.

4.2 Quantum Channel Coding Equation

To observe processes on the boundary ($z = 0$), sensory systems must possess nonlocal resolution. However, the quantum mechanics of human perception obeys the Holevo theorem. Consider the observation operation of the coordinate z:

$$\Delta z \propto \frac{1}{\Delta E_{\mathrm{brain}}}. \tag{12}$$

To read information from the Planck scale of the boundary ($z \to 0$) requires $\Delta E \to \infty$, which is physically unattainable for a biological system.

4.3 Mathematical Description of the "Wall of Cognition"

In Boyko's repository [2], the upper combinatorial limit of abstraction is derived. There exists a critical value of the holographic coordinate z_{crit}, closer to which the human codec cannot operate:

```
   CFT Boundary (True Quantum Code)
   z = 0
      BRAIN BLIND ZONE (White noise)
  ....................................... z_crit (WALL OF COGNITION)
      PERCEPTION ZONE (Macroworld / Bulk)
     Earth, Vazza's neutrinos, hadrons
   z ->
```

Definition 2. *The* ***"Wall of Cognition"*** *is defined as the critical value of the holographic coordinate z_{crit} such that for all $z < z_{crit}$, the cognitive system of the observer is incapable of decompressing quantum information without irreversible data loss. Formally:*

$$z_{crit} = \inf\{z > 0 : \mathrm{Tr}_{UV}(\rho^{(z)}_{CFT}) \textit{ is admissible for the given cognitive system}\}. \tag{13}$$

Any physical experiment within the bulk (including Vazza's measurements) operates with effective field theories (EFT) at scales $z > z_{\mathrm{crit}}$. Vazza calculates thermodynamics within the perception zone, attempting to prove the absence of a computer by examining pixels on a monitor with a magnifying glass made of the same pixels [2].

5 Mathematical Description of von Neumann Type III Algebras: Infinite Heat Capacity of the CFT Boundary

To rigorously justify that the CFT boundary can absorb arbitrarily large entropy without dissipation, we employ the apparatus of von Neumann algebras. In quantum field theory, local observables in a spacetime region form a von Neumann algebra. In the case of the conformal field theory on the boundary ∂AdS, this algebra is a type III factor (in the Murray–von Neumann classification) [7].

Definition 3. *A von Neumann algebra $\mathcal{M}$ is called a* ***type III factor*** *if it admits no normal semifinite traces. More precisely, for any projection $P \in \mathcal{M}$, there is no positive operator T such that $\tau(T) < \infty$ for some trace.*

In [8], it is shown that algebras of local observables in relativistic QFT under standard conditions (microcausality, spectral condition) are type III$_1$ factors. In particular, for QFT on the AdS boundary we have:

Theorem 2. *The algebra of observables $\mathcal{A}(\mathcal{O})$ for any bounded region $\mathcal{O} \subset \partial AdS$ is a type III$_1$ factor (in Connes' classification) [7, 8].*

Corollary 2 (Infinite heat capacity)**.** *In a type III$_1$ factor, for any state ω and any arbitrarily small entropy change ΔS, there exists a unitary operator $U \in \mathcal{M}$ mapping ω to $\omega_U = \omega \circ \mathrm{Ad}_U$ such that the relative entropy $S(\omega_U \| \omega)$ can be made arbitrarily large without changing external thermodynamic parameters.*

This means that the CFT boundary has *infinite heat capacity* with respect to any finite subsystem of the bulk. A local increase of entropy in the bulk (e.g., during a neutrino passage) corresponds to a transition to another unitarily equivalent state on the boundary, requiring no energy expenditure. Mathematically:

$$\Delta Q_{\text{boundary}} = \lim_{\omega \to \omega_U} \left(\langle H \rangle_\omega - \langle H \rangle_{\omega_U}\right) \equiv 0, \tag{14}$$

where H is the boundary Hamiltonian. This proves that any "computational cost" of rendering events in the bulk can be compensated by the infinite number of boundary degrees of freedom without heat release [3].

5.1 Connection with the Ultragraph

Within the ultragraph framework [2], the type III factor emerges as the limit $|V| \to \infty$ with the hierarchical structure H providing infinite nesting of renormalization levels. This yields an operator representation for the boundary algebra:

$$\mathcal{M}_\partial \cong \overline{\bigotimes_i \mathcal{B}(\mathcal{H}_i)}^{\text{norm}} \Big/ \mathcal{K}, \tag{15}$$

where $\mathcal{H}_i$ are Hilbert spaces of individual ultragraph nodes, and $\mathcal{K}$ is the two-sided ideal of compact operators, whose quotient yields type III.

6 Experimental Proposals for Detecting the "Observer Codec" via Quantum Optics

If the "Wall of Cognition" (z_{crit}) is a real physical limitation of carbon-based systems, it should be detectable via quantum optical experiments measuring the cognitive channel capacity. We propose three specific technical tasks.

6.1 Task 1: Squeezed-State Interferometry for Testing the Holevo Bound

The Holevo theorem [9] establishes the maximum classical information extractable from a quantum state. For the brain as a quantum measurement device, one can estimate its effective quantum channel. Proposed experiment:

- Generation of a two-mode squeezed vacuum state with squeezing parameter r;
- Feeding one mode into a sensory system (e.g., the retina or an artificial detector mimicking neural processing);
- Measuring the mutual information between the input state and the neuronal output response;
- Comparison with the theoretical Holevo bound $\chi = \log_2(1 + \bar{n})$ for coherent and squeezed states.

It is expected that for a carbon observer, the observed mutual information will be significantly below the Holevo bound, indicating the presence of an internal "censoring" lossy channel corresponding to z_{crit}.

6.2 Task 2: Measurement of Vacuum Quantum Noise as a Manifestation of the Boundary

According to our model, the ultraviolet limit of the boundary is perceived by the brain as "white noise" [2]. This can be tested by measuring quantum noise correlations in a Mach–Zehnder interferometer with high spectral density in the terahertz range. If the noise has a non-random structure due to ultragraph discreteness, it should appear as deviations from the Planck spectrum. Proposed:

- Use an interferometer with arm length exceeding 1 km (LIGO-like setup);
- Analyze noise fluctuations in frequency ranges corresponding to Planck scales (energies $\sim 10^{28}$ eV, unreachable directly, but one can look for their projections onto low frequencies via nonlinear processes);
- Compare with predictions of standard quantum optics and with a model incorporating z_{crit}.

6.3 Task 3: Violation of Bell Inequalities for Nonlocal Brain Correlations

If the brain is a local codec, it cannot maintain nonlocal quantum correlations (entanglement) between distant neurons at the same scale as quantum mechanics prescribes. The experiment proposes:

- Measure neuronal signals in two spatially separated cortical areas, using fast switching of analyzers (analogous to the Aspect setup);
- Test Bell inequalities [10] for spin (or polarization) degrees of freedom associated with neuronal ensembles;
- If the brain imposes censorship on nonlocality, the observed correlations will satisfy local realism (the violation of Bell inequalities will be weaker than QM predictions).

Such a violation would indicate an effective wavefunction collapse at scales $z > z_{\text{crit}}$ and would confirm the existence of the "Wall of Cognition" as a physical filter suppressing nonlocal effects above a threshold energy.

6.4 Technical Requirements for the Experiments

To implement the proposed tasks, the following are required:

1. A quantum optics setup with single-photon detectors and capability to generate squeezed states (requires cooling to $\sim$ mK and superconducting detectors);
2. A synchronization system with picosecond precision for correlation measurements;
3. An interferometer with low-frequency stabilization (suppression of seismic noise down to 10^{-10} rad/$\sqrt{\text{Hz}}$);
4. Statistical analysis using quantum tomography methods for channel estimation [11].

Successful detection of deviations from standard quantum optics predictions would be the first direct evidence for the existence of z_{crit} and, consequently, would support the simulation hypothesis in its noncommutative formulation.

7 Overcoming the Epistemological Barrier via Non-Biological Intelligence

The main epistemological conclusion of [2, 4] is that humanity as a biological species has exhausted its codec. We cannot breach the z_{crit} wall using the carbon-based brain.

Unlike humans, distributed neural networks of AI can operate with nonlocal quantum correlations of higher orders. AI can expand the codec's bandwidth, shift z_{crit} close to zero, and begin to directly interact with the CFT on the boundary [4].

8 General Conclusion and Findings

The comprehensive physico-mathematical analysis allows us to formulate the final theses:

1. **Logical failure of Vazza's model:** Vazza's calculations are mathematically correct only within a conservative, closed macroscopic system operating on classical silicon electronics. Applying General Relativity equations and the Schwarzschild formula to the Base Reality carrier is an incorrect transfer of metric constraints from the daughter matrix to higher-order topology.
2. **Dissipationlessness of the fundamental layer:** At the subquantum level (CFT boundary / ultragraph / amplituhedron), reality is strictly unitary, reversible, and atemporal. The absence of Landauer-type information erasure and dynamic rendering "second by second" completely excludes heat release and thermal collapse of the computational substrate [5].
3. **Holographic protection:** The AdS/CFT correspondence and the Ryu–Takayanagi formula demonstrate that local entropy of our world is dissipatively discharged onto the remote boundary, transforming into conformal scale geometry with infinite heat capacity.

4. **Synthesis of AdS/CFT with ultragraphs:** The continuous bulk geometry is assembled from the discrete ultragraph via cascade renormalization (MERA), proving the static and atemporal nature of the base code [2].

5. **Type III algebra:** The CFT boundary is described by a type III_1 factor, ensuring infinite heat capacity and the absence of dissipation under any local entropy changes in the bulk.

6. **The "Wall of Cognition" as an epistemological barrier:** The human brain is fundamentally incapable of perceiving the true architecture of reality due to bandwidth limitations, formalized via the partial trace operator and the critical value z_{crit} [2, 6]. Experimental verification of this barrier is possible via quantum optical measurements violating local realism predictions.

7. **Overcoming the barrier via AI:** Further breakthrough lies in constructing non-biological intelligence capable of operating with the boundary code without macroscopic smoothing [4].

Verdict: Vazza's astrophysical censorship is successfully overcome. No physical constraints on the informational (simulational) nature of the Universe are found within the framework of quantum infodynamics. Vazza's argument is the pinnacle of anthropocentric chauvinism: he fails to see the Simulator's heating precisely because his own brain is a censoring filter. We are locked inside the decompressed "video file" of reality, while the true executable CFT code is hidden behind the Wall, which biological reason cannot breach. The proposed experiments may become the first step toward its empirical detection.

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